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Testing for Granger-causality in quantiles

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ABSTRACT

This paper proposes a consistent parametric test of Granger-causality in quantiles. Although the concept of Granger-causality is defined in terms of the conditional distribution, most articles have tested Granger-causality using conditional mean regression models in which the causal relations are linear. Rather than focusing on a single part of the conditional distribution, we develop a test that evaluates nonlinear causalities and possible causal relations in all conditional quantiles, which provides a sufficient condition for Granger-causality when all quantiles are considered. The proposed test statistic has correct asymptotic size, is consistent against fixed alternatives, and has power against Pitman deviations from the null hypothesis. As the proposed test statistic is asymptotically nonpivotal, we tabulate critical values via a subsampling approach. We present Monte Carlo evidence and an application considering the causal relation between the gold price, the USD/GBP exchange rate, and the oil price.

KEYWORDS

Granger-causality; omnibus test; quantile regression; subsampling

JEL CLASSIFICATION

C12; C19; C22

1. Introduction

The Granger-causality definition proposed by Granger (1969) is the fundamental concept for studying dynamic relationships between time series. According to this definition, letting t denote time, a series Z_t is said to Granger-cause Y_{t+l} if Z_t incorporates information about the predictability for Y_{t+l} encompassed nowhere else in some large information set, which includes Y_{t-j} , $j \geq 0$. By using temporal precedence relationships, the notion of Granger-causality differs from the causal concepts based on controlled variation that are present in Haavelmo (1944), Holland (1986), Pearl (1988), Heckman and Vytlačil (2007), among others. Although time series Granger-causality tests lack the analysis of counterfactuals, they are straightforward to implement and do not suffer from the inferential problem of defining assumptions *a priori* required in a counterfactual analysis (Hoover, 2012). While the concept of Granger-causality is defined in terms of the conditional distribution, most articles have tested Granger-causality using conditional mean regression models in which the causal relations are linear. As a result, a conditional mean regression model cannot assess a tail causal relation or nonlinear causalities.

This article proposes a parametric omnibus test of Granger-causality in quantiles. The proposed approach allows us to evaluate nonlinear causalities, causal relations in conditional quantiles, and provides a sufficient condition for Granger-causality when all quantiles are considered. The quantile regression approach provides a more detailed and flexible analysis of the entire conditional distribution than the conditional mean-regression analysis, that focus only on a single part of the conditional distribution. In addition, a quantile causal relation may contrast with a causality in the mean of the conditional distribution. While a relationship with mean-causality shifts at least a non-negligible number of quantiles, a tail causal relation does not necessarily imply a causality in the mean. For example, Lee and Yang (2012) show that money-income Granger-causality in the conditional mean is quite weak and unstable, while it is significant in tail quantiles in most data sets. Finally, the proposed test is equivalent

to testing Granger-causality in distribution when all quantiles are considered. Rather than checking a necessary condition for Granger-causality, our approach analyzes a continuous space of conditional quantile functions that fully characterizes the concept of Granger-causality in distribution.

Our test is an application of the method proposed by Escanciano and Velasco (2010) in the context of Granger-causality. To our knowledge, testing for Granger-causality in quantiles by parametric methods in a flexible specification setting has not been analyzed in the literature before. Our test statistic is a Cramér-von Mises (CvM) functional norm of quantile-marked empirical processes that characterizes the null hypothesis of Granger noncausality. We reject the null hypothesis that past Z_t does not Granger-cause future Y_t whenever our test statistic is significantly different from zero, for any quantile over a continuum of quantile levels.

As the proposed test statistic is asymptotically nonpivotal and depends on the data generating process, we tabulate critical values via a subsampling method. The subsampling approach allows us to apply nonlinear conditional quantile regression models. Although our proposed test is computationally demanding, it has many interesting theoretical features: it does not require the choice of smoothing parameters, is consistent against all fixed alternatives, and is asymptotically strictly unbiased against a sequence of Pitman's local alternatives.

Chuang et al. (2009) and Yang et al. (2014) estimated the quantile causal effects by quantile regressions and tested the hypothesis of Granger noncausality by performing the Sup-Wald test of Koenker and Machado (1999) in all quantiles. We extend their method in two ways. First, our test provides an omnibus type of property: it requires only a model for the marginal quantile regression (under the null of no causality), and then searches for rejections of the null hypothesis in every direction, while the Sup-Wald test requires a particular model specification for the quantile regression under the alternative hypothesis of causality. In addition, we allow for nonlinear specifications of the quantile regression model under the null. Many causal relations are nonlinear, see for instance Bouezmarni et al. (2012). Therefore, a test based on a linear quantile regression model cannot be applied to testing nonlinear causality.

Regarding nonparametric approaches, Hong et al. (2009) proposed a nonparametric test of causality in Value-at-Risk (VaR), but their method provides only a necessary condition for Granger-causality. Jeong et al. (2012) extended the idea of Zheng (1998) to transform conditional quantile restrictions into conditional mean restrictions to testing causality in quantiles; more recently, Taamouti et al. (2014) proposed a nonparametric test for conditional density based Granger-causality. However, both testing procedures of the Granger-causality null hypothesis require beta-mixing conditions in the data generating process and are based on kernel methods. We provide two advantages. First, except for the application of the subsampling, our proposed causality test does not need mixing assumptions, and it requires only α -mixing assumptions for the validity of the subsampling, which are less restrictive than beta-mixing conditions. Moreover, our approach does not require the choice of smoothing parameters. Eventually, the misspecification of the parametric quantile regression model may affect the pattern of causality between the variables. As nonparametric estimation is model-free, nonparametric tests are robust to misspecification of the quantile regression. However, our parametric test is able to identify all the patterns of causality in the conditional distribution for flexible linear and nonlinear models, providing a clear interpretation of the causal relations. In sum, we believe that the proposed approach is a useful alternative to existing tests of Granger-causality in quantiles.

As further contributions, we investigate the finite sample performance of our method on simulated data, and we illustrate the empirical applicability of our setting by verifying the causal relation between the gold price, the USD/GBP exchange rate, and the oil price.

The rest of the article is organized as follows. In Section 2, we propose a test statistic for the null hypothesis of Granger noncausality in quantiles. In Section 3, we derive the asymptotic limit distribution of our test statistic under the null and the alternative hypotheses. We also prove that our test statistic has nontrivial power against $\sqrt{T}$ -local alternatives, with T the sample size, and we theoretically justify the validity of the subsampling approach in our framework. Section 4 presents Monte Carlo simulation results. In Section 5, we show an empirical application of our proposed test, and Section 6 concludes the article.

2. An omnibus test for Granger-causality in quantiles

2.1. Testing problem

Let $\{(Y_t, Z_t)\}_{t \in \mathbb{Z}}$ be a strictly stationary and ergodic time series process defined on some probability space $(\Omega, \mathcal{F}, P)$. Suppose $Y_t \in \mathbb{R}$ is the dependent variable and there is an explanatory vector $I_t \equiv (I_t^Y, I_t^Z)' \in \mathbb{R}^d$, $d = s + q$, where $I_t^Y := (Y_{t-1}, \dots, Y_{t-s})' \in \mathbb{R}^s$ and $I_t^Z := (Z_{t-1}, \dots, Z_{t-q})' \in \mathbb{R}^q$, for A' denoting the transpose matrix of A . Let $F_Y(y | I_t^Y, I_t^Z)$ and $F_Y(y | I_t^Y)$ be the conditional distribution functions of Y_t given (I_t^Y, I_t^Z) and I_t^Y , respectively. We assume they are continuous for all $y \in \mathbb{R}$.

According to Granger (1969), a series Z_t does not Granger-cause another series Y_t if past Z_t does not help to predict future Y_t given past Y_t . We characterize the null hypothesis of Granger noncausality from Z_t to Y_t as follows:

$$H_0^{Z \rightarrow Y} : F_Y(y | I_t^Y, I_t^Z) = F_Y(y | I_t^Y), \quad \text{for all } y \in \mathbb{R}. \quad (1)$$

We denote the null hypothesis of (1) as Granger noncausality in distribution. Since the estimation of the conditional distribution may be complicated in practice, many articles have tested Granger noncausality in mean, which is only a necessary condition for (1). In this case, Z_t does not Granger cause Y_t in mean if

$$E(Y_t | I_t^Y, I_t^Z) = E(Y_t | I_t^Y) \text{ a.s.}, \quad (2)$$

where $E(Y_t | I_t^Y, I_t^Z)$ and $E(Y_t | I_t^Y)$ denote the mean of $F_Y(\cdot | I_t^Y, I_t^Z)$ and $F_Y(\cdot | I_t^Y)$, respectively. Granger noncausality in mean of (2) can be easily extended to higher order moments, see for example Cheung and Ng (1996). However, causality in mean (or in higher moments) overlooks the dependence that may appear in conditional tails of the distribution. On the other hand, the Granger noncausality distribution of (1) does not inform us about the level where the causality exists, if (1) is rejected. Thus, we propose to test Granger noncausality in conditional quantiles, since it allows us to determine the pattern of causality and it provides a sufficient condition for testing Granger noncausality in distribution of (1), as the quantiles completely characterize a distribution. Let $Q_\tau^{Y,Z}(\cdot | I_t^Y, I_t^Z)$ and $Q_\tau^Y(\cdot | I_t^Y)$ denote the τ -quantiles of $F_Y(\cdot | I_t^Y, I_t^Z)$ and $F_Y(\cdot | I_t^Y)$ respectively. We can equally test (1) as

$$H_0^{QC:Z \rightarrow Y} : Q_\tau^{Y,Z}(Y_t | I_t^Y, I_t^Z) = Q_\tau^Y(Y_t | I_t^Y), \text{ a.s. for all } \tau \in \mathcal{T}, \quad (3)$$

where $\mathcal{T}$ is a compact set such that $\mathcal{T} \subset [0, 1]$ and the conditional τ -quantiles of Y_t satisfy the following restrictions:

$$\begin{aligned} \Pr\{Y_t \leq Q_\tau^Y(Y_t | I_t^Y) | I_t^Y\} &:= \tau, \text{ a.s. for all } \tau \in \mathcal{T}, \\ \Pr\{Y_t \leq Q_\tau^{Y,Z}(Y_t | I_t^Y, I_t^Z) | I_t^Y, I_t^Z\} &:= \tau, \text{ a.s. for all } \tau \in \mathcal{T}. \end{aligned} \quad (4)$$

Given an explanatory vector I_t , we have $\Pr\{Y_t \leq Q_\tau(Y_t | I_t) | I_t\} = E\{\mathbb{1}[Y_t \leq Q_\tau(Y_t | I_t)] | I_t\}$, where $\mathbb{1}(a \leq b)$ is an indicator function of the event that a is less or equal than b . Thus (3) is equivalent to

$$E\left\{\mathbb{1}[Y_t \leq Q_\tau^{Y,Z}(Y_t | I_t^Y, I_t^Z)] \mid I_t^Y, I_t^Z\right\} = E\left\{\mathbb{1}[Y_t \leq Q_\tau^Y(Y_t | I_t^Y)] \mid I_t^Y, I_t^Z\right\}, \quad \text{a.s. for all } \tau \in \mathcal{T}, \quad (5)$$

where the left-hand side of (5) is equal to the τ -quantile of $F_Y(\cdot | I_t^Y, I_t^Z)$ by definition. We postulate a parametric model to estimate the τ th quantile of $F_Y(\cdot | I_t)$, where we assume that $Q_\tau(\cdot | I_t)$ is correctly specified by a parametric model $m(\cdot, \theta(\tau))$ belonging to a family of functions $\mathcal{M} = \{m(\cdot, \theta(\tau)) | \theta(\cdot) : \tau \mapsto \theta(\tau) \in \Theta \subset \mathbb{R}^p, \text{ for } \tau \in \mathcal{T} \subset [0, 1]\}$. Let $\mathcal{B} \subset \mathcal{M}$ be a family of uniformly bounded functions $\tau \mapsto \theta(\tau)$ such that $\theta(\tau) \in \Theta \subset \mathbb{R}^p$. Then, under the null hypothesis in (3), the τ -conditional quantile $Q_\tau^Y(\cdot | I_t^Y)$ is correctly specified by a parametric model $m(I_t^Y, \theta_0(\tau))$, for some $\theta_0 \in \mathcal{B}$, using only the restricted information set I_t^Y , and we redefine our testing problem in (3) as

$$H_0^{Z \rightarrow Y} : E\left[\mathbb{1}(Y_t \leq m(I_t^Y, \theta_0(\tau))) \mid I_t^Y, I_t^Z\right] = \tau, \text{ a.s. for all } \tau \in \mathcal{T}, \quad (6)$$

versus

$$\mathcal{H}_A^{Z \rightsquigarrow Y} : E[\mathbb{1}(Y_t \leq m(I_t^Y, \theta_0(\tau))) | I_t^Y, I_t^Z] \neq \tau, \text{ for some } \tau \in \mathcal{T}, \quad (7)$$

where $m(I_t^Y, \theta_0(\tau))$ is the only element of $\mathcal{M}$ that correctly specifies the true conditional quantile $Q_\tau^Y(\cdot | I_t^Y)$, for all $\tau \in \mathcal{T}$. We rewrite (6) as $\mathcal{H}_0^{Z \rightsquigarrow Y} : E[\Psi_{\tau,t}(\theta_0) | I_t^Y, I_t^Z] = 0$ almost surely, for all $\tau \in \mathcal{T}$, to simplify notation, where $\Psi_{\tau,t}(\theta_0) \equiv \Psi_\tau(Y_t - m(I_t^Y, \theta_0(\tau)))$ for a function $\Psi_\tau(\varepsilon) := \mathbb{1}(\varepsilon \leq 0) - \tau$.

By appropriately choosing a parametric family of functions $\{g(I_t, \omega) : \omega \in \mathcal{W} \subset \mathbb{R}^r\}$, where $I_t = (I_t^Y, I_t^Z)'$ and $\mathcal{W}$ is a compact subset of $\mathbb{R}^r$, for $r \leq d$, we may restate the null hypothesis (6) as a sequence of unconditional moment restrictions $E[\Psi_{\tau,t}(\theta_0)g(I_t, \omega)] = 0$, for all $\tau \in \mathcal{T}$ and for all measurable functions $g(I_t, \omega)$ such that $E[|g(I_t, \omega)|] < \infty$ (see Bierens, 1982; Escanciano, 2006; Stinchcombe and White, 1998). For example, two possible choices of weighting functions that satisfy the previous equivalence are $g(I_t, \omega) = \mathbb{1}(I_t \leq \omega)$, for $\omega \in [-\infty, \infty]^d$, and $g(I_t, \omega) = \exp(i\omega' I_t)$, for $\omega \in \mathbb{R}^d$, where $i = \sqrt{-1}$ is the imaginary root. We could also use any parametric family of weighting functions proposed in Kuan and Lee (2004) and Escanciano (2006). However, unreported simulations suggested that the weighting function $g(I_t, \omega) := \exp(i\omega' I_t) = \exp[i(\omega_1(Y_{t-1}, Z_{t-1})' + \dots + \omega_r(Y_{t-r}, Z_{t-r})')]$, for all $\omega \in \mathbb{R}^r$ with $r \leq d$, outperforms in terms of size and power other alternative parametric family of functions. Therefore, by Lemma 1 of Escanciano (2006), we characterize the null hypothesis (6) by the following infinite set of unconditional moment restrictions

$$E[\Psi_{\tau,t}(\theta_0) \exp(i\omega' I_t)] = 0, \quad \text{for all } \omega \in \mathbb{R}^d \text{ and for all } \tau \in \mathcal{T}, \quad (8)$$

where $i = \sqrt{-1}$ is the imaginary root. Our test statistic is based on the sample analog of the moment restriction of (8)

$$v_T(\omega, \tau) := \frac{1}{\sqrt{T}} \sum_{t=1}^T \Psi_{\tau,t}(\theta_T) \exp(i\omega' I_t), \quad (9)$$

where $\theta_T(\tau)$ is a $\sqrt{T}$ -consistent estimator of $\theta_0(\tau)$, for all $\tau \in \mathcal{T}$. Our proposed framework allows for any $\sqrt{T}$ -consistent estimator of $\theta_T(\tau)$ satisfying some mild conditions (described in the next section) such as the quantile regression estimator by Koenker and Bassett (1978), the quantile autoregressive estimator by Koenker and Xiao (2006), and the CAViaR estimator by Engle and Manganelli (2004). Given our sample $\{(Y_t, Z_t)\}_{t \in \mathbb{Z}}$, we define $v_T(\omega, \tau)$ as the quantile marked-residual process, indexed by $\omega \in \mathbb{R}^d$ and $\tau \in \mathcal{T}$. Our proposed test statistic S_T is a CvM functional norm of $v_T(\omega, \tau)$ defined as

$$S_T := \int_{\mathcal{T}} \int_{\mathcal{W}} |v_T(\omega, \tau)|^2 dF_\omega(\omega) dF_\tau(\tau), \quad (10)$$

where $F_\omega(\cdot)$ and $F_\tau(\cdot)$ are some integrating measures on $\mathcal{W}$ and $\mathcal{T}$, respectively, and $\mathcal{W}$ is a compact subset of $\mathbb{R}^d$ containing the origin. Following Epps and Pulley (1983), three considerations influence the choice of $F_\omega(\cdot)$. The first is that it should assign high weight where $|v_T(\omega, \tau)|$ is large under the alternative hypothesis. The second is that $F_\omega(\cdot)$ should give high weight where the statistic $v_T(\omega, \tau)$ is a relatively precise estimator of $E[\Psi_{\tau,t}(\theta_0) \exp(i\omega' I_t)]$. It can be shown that this precision is greatest near the origin (Epps and Pulley, 1983). Thus, it seems that $F_\omega(\cdot)$ should assign high weight in some interval around the origin. A final practical consideration is that $F_\omega(\cdot)$ be such that S_T in (10) have a closed form. For example, the normal or exponential cumulative distribution functions (CDF) are weighting functions that satisfy these conditions (Kuan and Lee, 2004). Following Escanciano and Velasco (2006), we take $F_\omega(\cdot)$ as the CDF of a d -variate standard normal random vector. This choice simplifies the computation of S_T , avoiding numerical integration, and satisfies the other two requirements of the weighting function. Hong (1999) shows that the power performance for similar tests to S_T are not too sensitive to the choice of $F_\omega(\cdot)$.

Once we take $F_{\omega}(\cdot)$ as the CDF of a standard normal random variable, the vector of weights $\omega \in \mathbb{R}^d$ is drawn from a standard normal distribution and the test statistic in (10) can be estimated using its sample counterpart. We choose $F_{\tau}(\cdot)$ as a uniform discrete distribution over a grid of $\mathcal{T}$ in n equidistributed points, $\mathcal{T}_n = \{\tau_j\}_{j=1}^n$. Let Ψ be the $T \times n$ matrix Ψ with elements $\psi_{i,j} = \Psi_{\tau_j}(Y_i - m(\mathbf{I}_i^Y, \theta_T(\tau_j)))$. Then, our test statistic S_T has the form

$$S_T = \frac{1}{Tn} \sum_{j=1}^n \left| \psi'_{\cdot j} \mathbf{W} \psi_{\cdot j} \right|, \quad (11)$$

where $\mathbf{W}$ is the $T \times T$ matrix with elements $\mathbf{w}_{t,s} = \exp[-0.5(\mathbf{I}_t - \mathbf{I}_s)^2]$, and $\psi_{\cdot j}$ denotes the j th column of Ψ , see Kuan and Lee (2004) and Escanciano and Velasco (2006) for more details. We choose the CvM functional norm because unreported simulations suggested that the CvM type statistics provide better size and power results than the ones implied by other continuous functional norms such as the Kolmogorov–Smirnov functional norm. Under the assumptions described in the next section, the test statistic S_T weakly converges to zero under the null hypothesis (6), and to a probability limit different than zero under the alternative (7). We reject the null hypothesis whenever we observe “large” values of S_T .

2.2. Subsampling critical values

As the null distribution of test statistic S_T is asymptotically nonpivotal and depends on the data generating process (DGP), we propose a subsampling procedure to calculate critical values for S_T of (10). Horowitz (1998) and Sakov and Bickel (2000), among others, have proposed resampling methods for quantile regression models, but their approach is valid only for independent and identically distributed (i.i.d.) data. Subsampling is a resampling method that provides asymptotic inference for general quantile regression models in a time series context, see Politis et al. (1999) and Chernozhukov and Fernández-Val (2005). Under the subsampling method, critical values are obtained by mimicking the test statistic S_T of (10).

Since the asymptotic properties of the subsampling tests depend on the choice of the subsample size b , there are different rules proposed in the literature for choosing the subsample size (see Politis et al., 1999; Sakov and Bickel, 2000). Whereas Politis et al. (1999) derived calibration and minimum volatility methods for picking the optimal subsample size, Sakov and Bickel (2000) suggested choosing a subsample size $b = \lfloor kT^{2/5} \rfloor$, where $\lfloor \cdot \rfloor$ is the integer part of a number, k is a constant parameter, and T is the size of sample. Our simulations indicated that the subsample size proposed in Sakov and Bickel (2000) works well in our framework. Therefore, we follow the approach of Sakov and Bickel (2000) and choose a subsample of size $b = \lfloor kT^{2/5} \rfloor$, for different values of k . Section 4 shows that the subsampling statistics yields good inference for different choices of the constant k .

Although some authors recommend the use of recentering in subsampling (see, e.g., Chernozhukov and Fernández-Val, 2005), unreported simulations suggest that the noncentered subsampling test statistic performs better than the recentered one, and thus we omit it. To describe the calculation of the subsampling critical values, we write the empirical process $v_T(\omega; \tau)$ for the test statistic as a function of the data $\{X_t = (Y_t, Z_t)\}_{t \in \mathbb{Z}}$, $v_T(\omega; \tau) = v_T(X_1, \dots, X_T; \tau)$. We describe the algorithm for computing a subsampling realization of our test statistic S_T in two steps.

1. Given a time series $\{X_t = (Y_t, Z_t)\}$ of sample size T , construct $B = T - b + 1$ subsamples of size b of the form $\{X_i, \dots, X_{i+b-1}\}$. For each i th subsample, $i \leq B$, compute the inference process $v_{b,i}(X_i, \dots, X_{i+b-1}; \tau)$. Let $v_T(\omega; \tau) \equiv v_T(X_1, \dots, X_T; \tau)$ be the inference process (9) computed over the entire sample T ; denote by $v_{b,i}(\omega; \tau) \equiv v_{b,i}(X_i, \dots, X_{i+b-1}; \tau)$ the inference process computed over the i th subsample of data, and define

$$S_{b,i} \equiv \int_{\mathcal{T}} \int_{\mathcal{W}} |v_{b,i}(\omega, \tau)|^2 dF_{\omega}(\omega) dF_{\tau}(\tau),$$

where $S_{b,i}$ is our test statistic computed with the i th subsample $\{X_i, \dots, X_{i+b-1}\}$ of size b .

2. As each subsample of size b is taken without replacement from the original data, the subsample b is indeed a sample of size b from the true model. Define $G(x) \equiv \Pr(S_T \leq x)$ as the CDF of our test statistic. Estimate $G(x)$ by $\hat{G}(x) = B^{-1} \sum_{i=1}^B \mathbb{1}(S_{b,i} \leq x)$. Obtain the critical values as the $(1 - \tau)$ th quantile of $\hat{G}$, that is $c_{T,b}(1 - \tau) = \hat{G}^{-1}(1 - \tau)$. Finally, reject the null hypothesis (6) if $S_T > c_{T,b}(1 - \tau)$.

As we are dealing with quantile regression models, we implement the subsampling above over a grid of quantiles $\mathcal{T}_n = \{\tau_j\}_{j=1}^n$ in n equidistributed points in place of $\mathcal{T}$. The asymptotic theory of the subsampling test statistic allows for $n \rightarrow \infty$ as the sample size T grows to infinity and the grid $\mathcal{T}_n$ generated independently from a distribution over $\mathcal{T}$, but for practical considerations we considered a deterministic grid $\mathcal{T}_n$ with fixed n in this article.

3. Asymptotic theory

In this section, we derive the asymptotic distributions of our test statistic S_T under the null and alternative hypothesis. We consider the process $v_T(\omega, \tau)$ of (9) as a mapping from $(\Omega, \mathcal{F}, P)$ taking values in $\ell^\infty(\mathcal{W} \times \mathcal{T})$, that is the set of all complex-valued uniformly bounded functions defined with the supremum metric, d_∞ , and $\mathcal{B}_{d_\infty}$ is its Borel σ -algebra. Hereafter “ $\Rightarrow$ ” denotes the weak convergence on $(\mathcal{B}_{d_\infty}, d_\infty)$, and C is a fixed constant. Let $\mathcal{F}_t = \sigma(\mathbf{I}_t^Y, \mathbf{I}_{t-1}^Y, \dots)$ be the σ -field generated up to time t . We define the τ -quantile innovation, for each $t \in \mathbb{Z}$, as $\varepsilon_t(\tau) := Y_t - Q_\tau(\mathbf{I}_t^Y)$ and the parametric quantile error as $e_t(\theta(\tau)) := Y_t - m(\mathbf{I}_t^Y, \theta(\tau))$. In addition, f_x denotes the density function of a conditional distribution function F_x . All limits are taken as $T \rightarrow \infty$, where T is the sample size. We maintain the following main assumptions to analyze the asymptotic behavior of our test statistic.

Assumption 1. $\{(Y_t, Z_t)\}_{t \in \mathbb{Z}}$ is a strictly stationary and ergodic process, with $E[|\mathbf{I}_0^Y|^2] < C$. Under $\mathcal{H}_0^{Z \rightarrow Y}$ of (6), $\{\Psi_{\tau,t}(\theta_0(\tau)), \mathcal{F}_t\}$ is a martingale difference sequence for all $\tau \in \mathcal{T}$. The parametric family $m(\cdot, \theta_0(\tau))$ is nondecreasing in τ a.s. The family of distributions functions $\{F_x, x \in \mathbb{R}^d\}$ has Lebesgue densities $\{f_x, x \in \mathbb{R}^d\}$ that are equicontinuous and uniformly bounded away from zero for the quantiles of interest.

Assumption 2. For each $\theta_1 \in \mathcal{B}$, we have as follows:

(a) There exists a vector of functions $g_{t-1} : \Theta \mapsto \mathbb{R}^d$, for $g_{t-1}(\theta_1(\tau))$ $\mathcal{F}_{t-1}$ -measurable for each $t \in \mathbb{Z}$ satisfying, for all $k < \infty$,

$$\sup_{1 \leq t \leq T, \|\theta_1 - \theta_2\|_{\mathcal{B}} \leq kT^{-1/2}} T^{1/2} \|m(\mathbf{I}_t^Y, \theta_2) - m(\mathbf{I}_t^Y, \theta_1) - (\theta_2 - \theta_1)' g_{t-1}(\theta_1)\|_{\mathcal{B}} = o_p(1).$$

(b) For all sufficiently small $\delta > 0$,

$$E \left[\sup_{\|\theta_1 - \theta_2\|_{\mathcal{B}} \leq \delta} |\mathbb{1}(Y_t \leq m(\mathbf{I}_t^Y, \theta_1(\tau))) - \mathbb{1}(Y_t \leq m(\mathbf{I}_t^Y, \theta_2(\tau)))| \right] \leq C\delta, \text{ for all } \tau \in \mathcal{T},$$

and

$$E \left[\sup_{|\tau_1 - \tau_2| \leq \delta} |m(\mathbf{I}_t^Y, \theta_1(\tau_1)) - m(\mathbf{I}_t^Y, \theta_1(\tau_2))| \right] \leq C\delta.$$

(c) Uniformly in $\tau \in \mathcal{T}$, $E|g_{t-1}(\theta_1(\tau))|^2 < \infty$, and uniformly in $(x', \tau)' \in \mathcal{W} \times \mathcal{T}$,

$$\begin{aligned} & \left| \frac{1}{T} \sum_{t=1}^T g_{t-1}(\theta_0(\tau)) \exp(i\omega' \mathbf{I}_t) f_{\mathbf{I}_t^Y}(m(\mathbf{I}_t^Y, \theta_0)) \right. \\ & \quad \left. - E[g_{t-1}(\theta_0(\tau)) \exp(i\omega' \mathbf{I}_t) f_{\mathbf{I}_t^Y}(m(\mathbf{I}_t^Y, \theta_0))] \right| = o_p(1). \end{aligned}$$

Assumption 3. Let $N_{[\cdot]}(\delta, \mathcal{G}, \|\cdot\|_{\mathcal{B}})$ be the δ -bracketing number of a class of functions $\mathcal{G}$ with respect to a norm $\|\cdot\|$. The parametric space Θ is compact in $\mathbb{R}^P$. The true parameter $\theta_0(\tau)$ belongs to the interior of Θ for each $\tau \in \mathcal{T}$, and $\theta_0 \in \mathcal{B}$. The class $\mathcal{B}$ satisfies

$$\int_0^\infty (\log(N_{[\cdot]}(\delta^2, \mathcal{B}, \|\cdot\|_{\mathcal{B}})))^{1/2} d\delta < \infty.$$

Assumption 4. The estimator θ_T satisfies that $\Pr(\theta_T \in \mathcal{B}) \rightarrow 1$ as $T \rightarrow \infty$, and the asymptotic expansion under $\mathcal{H}_0^{Z \rightarrow Y}$ of (6), uniformly in $\tau \in \mathcal{T}$,

$$Q_T(\tau) = \sqrt{T}(\theta_T(\tau) - \theta_0(\tau)) = \frac{1}{\sqrt{T}} \sum_{t=1}^T \ell_\tau(Y_t, \mathbf{I}_t^Y, \theta_0(\tau)) + o_p(1),$$

where $E[\ell_\tau(Y_1, \mathbf{I}_0^Y, \theta_0(\tau))] = 0$, $E[\ell_\tau(Y_1, \mathbf{I}_0^Y, \theta_0(\tau)) \ell'_\tau(Y_1, \mathbf{I}_0^Y, \theta_0(\tau))]$ exists, and it is positive definite, and $E[\ell_\tau(Y_t, \mathbf{I}_t^Y, \theta_0(\tau)) \Psi_{\tau,s}(\theta_0)] = 0$ if $t \neq s$. As a process in $\ell^\infty(\mathcal{T})$, $Q_T(\tau)$ converges weakly to a Gaussian process $Q(\cdot)$ with zero mean and covariance function

$$K_Q(\tau_1, \tau_2) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T E[\ell_{\tau_1}(Y_t, \mathbf{I}_t^Y, \theta_0(\tau_1)) \times \ell_{\tau_2}(Y_s, \mathbf{I}_s^Y, \theta_0(\tau_2))].$$

Assumption 5. Under the alternative hypothesis $\mathcal{H}_A^{Z \rightarrow Y}$ of (7) we have as follows:

- (a) There exists a $\theta_1 \in \mathcal{B}$ such that $\|\theta_T - \theta_1\|_{\mathcal{B}} = o_p(1)$;
- (b) $E\{\Psi_{\tau,t}(\theta_0) \exp(i\omega' \mathbf{I}_t)\} \neq 0$ in a subset with positive Lebesgue measure on $\mathcal{W} \times \mathcal{T}$.

Assumptions 1–5 are very similar to Assumptions A1–A5 of Escanciano and Velasco (2010), but we consider most of them in a context of a restricted information set $\mathbf{I}_t^Y$ rather than in the full one $\mathbf{I}_t \equiv (\mathbf{I}_t^Y, \mathbf{I}_t^Z)$. Assumptions 1–3 provide conditions for the validity of a functional central limit theorem for empirical processes in the dependent data case. Assumption 4 is required to guarantee the asymptotic distribution of our test when nonlinear quantile regression models are applied, see for example Mukherjee (1999). Assumption 5 provides a sufficient condition for the estimator θ_T to be consistent under a fixed alternative hypothesis, see Angrist et al. (2006) for conditions to satisfy Assumption 5(a); Assumption 5(b) holds if $\mathbf{I}_t$ is bounded. Theorem 1 below is an application of Theorems 1–3 of Escanciano and Velasco (2010) and the continuous mapping theorem.

Theorem 1. Under Assumptions 1–5, we have as follows:

- (i) Under the null hypothesis $\mathcal{H}_0^{Z \rightarrow Y}$ of (6),

$$S_T \xrightarrow{d} \int_{\mathcal{T}} \int_{\mathcal{W}} |\mathbb{G}_1(\omega, \tau)|^2 dF_{\omega}(\omega) dF_{\tau}(\tau),$$

where $\mathbb{G}_1$ is a tight mean zero Gaussian process.

- (ii) Under the alternative hypothesis $\mathcal{H}_A^{Z \rightarrow Y}$ of (7), there exists an $\varepsilon > 0$ such that

$$\lim_{T \rightarrow \infty} \Pr(S_T > \varepsilon) = 1.$$

Theorem 1 shows that the asymptotic null distribution of S_T is a functional of a zero-mean Gaussian process $\mathbb{G}_1$. By Theorem 1, we reject the null hypothesis $\mathcal{H}_0^{Z \rightarrow Y}$ whenever S_T is significantly positive.

3.1. Local alternatives and subsampling validity

Now we analyze the asymptotic distribution of S_T against a sequence of Pitman's local alternatives converging to the null hypothesis at rate $\sqrt{T}$, where T denotes the sample size. Under a sequence of

local alternatives $\mathcal{H}_{A,T}^{Z \rightsquigarrow Y}$, we have

$$\mathcal{H}_{A,T}^{Z \rightsquigarrow Y} : E[\Psi_{\tau,t}(\theta_0) | \mathbf{I}_t^Y, \mathbf{I}_t^Z] = \delta_\tau / \sqrt{T}, \text{ a.s. for all } \tau \in \mathcal{T}, \quad (12)$$

where δ_τ is a function satisfying the following assumption.

Assumption 6. The function $\delta_\tau : \mathbb{R}^d \mapsto \mathbb{R}$ has the following properties:

- (i) $E\{\sup_{\tau \in \mathcal{T}} |\delta_\tau(\mathbf{I}_t^Y)|\} < \infty$;
- (ii) There exists a $\mathbf{I}_t^Y$ -measurable random variable h_{t-1} with $E[h_{t-1}^2] < \infty$ such that, for all $t \in \mathbb{Z}$ and for all $\tau_1, \tau_2 \in \mathcal{T}$, $|\delta_{\tau_1}(\mathbf{I}_t^Y) - \delta_{\tau_2}(\mathbf{I}_t^Y)| \leq h_{t-1} |\tau_1 - \tau_2|$, a.s.

Assumption 6 is analogous to Assumption A6 of Escanciano and Velasco (2010), but we consider just the restricted information set $\mathbf{I}_t^Y$. To ensure nontrivial local power of our proposed test, we need to impose an assumption on the estimator θ_T under a local alternative as given in (12). Then we modify Assumption 4 as follows.

Assumption 4'. Under a local alternative $\mathcal{H}_{A,T}^{Z \rightsquigarrow Y}$ in (12),

$$\sqrt{T}(\theta_T(\tau) - \theta_0(\tau)) = \eta_a(\tau) + \frac{1}{\sqrt{T}} \sum_{t=1}^T \ell_\tau(Y_t, \mathbf{I}_t^Y, \theta_0(\tau)) + o_p(1),$$

uniformly in τ , where ℓ_τ satisfies the same conditions as in Assumption 4 and $\eta_a(\tau) \in \mathbb{R}^p$ for each $\tau \in \mathcal{T}$.

Assumption 4' can be applied to most quantile regression estimators in the literature, see for example Mukherjee (1999). Theorem 2 demonstrates that under a local alternative $\mathcal{H}_{A,T}^{Z \rightsquigarrow Y}$ of (12) the asymptotic distribution of S_T has an extra shift function implying consistency against $\sqrt{T}$ -alternatives. Theorem 2 follows from Theorem 4 of Escanciano and Velasco (2010) and the continuous mapping theorem.

Theorem 2. Under the local alternatives $\mathcal{H}_{A,n}^{QC}$ in (12), Assumptions 1–3, 4', and 5, we have

$$S_T \xrightarrow{d} \int_{\mathcal{T}} \int_{\mathcal{W}} |\mathbb{G}_1(\omega, \tau) + \Delta(\omega, \tau)|^2 dF_\omega(\omega) dF_\tau(\tau),$$

where $\Delta(\omega, \tau)$ is a nontrivial shift function.

Now we derive the asymptotic validity of the subsampling critical values described in Section 2.2. Although no mixing conditions are required for the convergence of S_T , we need another assumption on the serial dependence of the data generating process to validate the subsampling theoretically. According to Politis et al. (1999), Assumption 7 is sufficient for the asymptotic validity of the critical values generated by the subsampling approach.

Assumption 7. $\{(Y_t, Z_{t+1})'\}_{t \in \mathbb{Z}}$ is a strictly stationary strong mixing process with α -mixing coefficients satisfying $\sum_{i=1}^T \alpha(i)/T \rightarrow 0$ as $T \rightarrow \infty$.

The next result delivers the asymptotic distribution of the subsampling test statistic. Since it is an application of Theorem 2 of Whang (2006) and Theorem 5 of Escanciano and Velasco (2010), we omit the proof.

Theorem 3. Under Assumptions 1–7, $b/T \rightarrow 0$, and $b \rightarrow \infty$ as $T \rightarrow \infty$, we have as follows:

- (i) Under the null hypothesis $\mathcal{H}_0^{Z \rightsquigarrow Y}$ in (6),

$$\lim_{T \rightarrow \infty} \Pr(S_T > c_{T,b}(1 - \tau)) = \tau.$$

(ii) Under the fixed alternative hypothesis $\mathcal{H}_A^{Z \rightarrow Y}$ in (7),

$$\lim_{T \rightarrow \infty} \Pr(S_T > c_{T,b}(1 - \tau)) = 1.$$

(iii) Under the local alternative $\mathcal{H}_{A,T}^{Z \rightarrow Y}$ in (12),

$$\lim_{T \rightarrow \infty} \Pr(S_T > c_{T,b}(1 - \tau)) \geq \tau.$$

Theorem 3 shows that our test based on the subsampling critical value has asymptotically correct size, is consistent, and is able to detect alternatives tending to the null at the parametric rate $\sqrt{T}$.

4. Monte Carlo experiments

In this section, we perform Monte Carlo simulation experiments with DGPs under the null and the alternative hypothesis to evaluate the finite-sample performance of our proposed test statistic. The data are generated from the following DGPs:

$$\text{DGP1: } Y_t = 0.5Y_{t-1} + cZ_{t-1} + \varepsilon_{1t}, \text{ and } Z_t = \varepsilon_{2t}, \quad (13)$$

$$\text{DGP2: } Y_t = 0.5Y_{t-1} + cZ_{t-1} + \varepsilon_{1t}, \text{ and } Z_t = 1 + 0.8Z_{t-1} + \varepsilon_{2t}, \quad (14)$$

$$\text{DGP3: } Y_t = 0.5Y_{t-1} + cZ_{t-1}^2 + \varepsilon_{1t}, \text{ and } Z_t = 1 + 0.8Z_{t-1} + \varepsilon_{2t}, \quad (15)$$

where $\varepsilon_{it} \sim \text{i.i.d. } N(0, 1)$, for $i = 1, 2$. For all DGPs above, we have under the null hypothesis $c = 0.00$, where the coefficient c captures the degree of causality from past Z_t to future Y_t , and hence a higher absolute value of c implies a stronger Granger-causality. The coefficients of the DGPs above assure that the generated time series are stationary. To see how the rejection frequencies change in relation to the ratios of the coefficients of Y_{t-1} and Z_{t-1} to the variance of the error term, we design another DGP where these two ratios increase with the value of c . Fixing the value of the coefficient of Y_{t-1} to 0.4 and the variance of the error term to $(1 - 0.4 - c) = (0.6 - c)$, then the ratios of the coefficients of Y_{t-1} and Z_{t-1} to the variance of the error term ε_{3t} become higher as c increases:

$$\text{DGP4: } Y_t = 0.4Y_{t-1} + cZ_{t-1} + \varepsilon_{3t}, \text{ and } Z_t = \varepsilon_{2t} \quad (16)$$

where $\varepsilon_{3t} \sim \text{i.i.d. } N(0, (0.6 - c))$. We consider the empirical rejection frequencies for 5% nominal level tests for different sample sizes T , subsample sizes b , conditional quantile parametric models $m(\cdot, \cdot)$, and causality parameters c . We consider the sample sizes $T = 100$, $T = 250$, and $T = 500$. To show that our test is robust to the choice of the subsample size, we use three different subsample sizes $b = \lfloor kT^{2/5} \rfloor$ for each sample T , where $\lfloor \cdot \rfloor$ is the integer part of a number, with $k = \{3, 4, 5\}$. Thus for $k = 3, 4$, and 5, we have $b = 18, 25$, and 31 for $T = 100$, $b = 27, 36$, and 45 for $T = 250$, and $b = 36, 48$, and 60 for $T = 500$. We propose two different parametric quantile auto-regressive specifications $m(\cdot)$ —quantile AR(1) and AR(2)—for modeling the conditional quantiles of Y_t , for all $\tau \in \mathcal{T}$, as follows:

$$\begin{aligned} m^1(I_t^Y, \theta_T(\tau)) &= \mu_0(\tau) + \mu_1(\tau)Y_{t-1} + \sigma_t\Phi_\varepsilon^{-1}(\tau), \\ m^2(I_t^Y, \theta_T(\tau)) &= \mu_0(\tau) + \mu_1(\tau)Y_{t-1} + \mu_2(\tau)Y_{t-2} + \sigma_t\Phi_\varepsilon^{-1}(\tau), \end{aligned} \quad (17)$$

where the parameters $\theta_T(\tau) = (\mu_0(\tau), \mu_1(\tau), \mu_2(\tau), \sigma_t)'$ are estimated by maximum likelihood in an equally spaced grid of 20 quantiles on the interval $\mathcal{T} = [0.10, 0.90]$. When $c = 0.00$, there is no causality from past values of Z_t to Y_t and the rejection rates deliver the empirical size of our test statistic. For $c \neq 0.00$, there is causality from lagged values of Z_t to Y_t and the rejection rates yield the empirical power of our test statistic. We apply 1,000 Monte Carlo replications in each of the simulations, which implies a maximal simulation standard error for the empirical sizes and powers of the test of $\max_p \sqrt{p(1-p)/1000} \approx 0.016$.

Figures 1 and 2 show the rejection frequencies of the S_T test¹ using a QAR(1) and a QAR(2) model of (17), respectively. The proposed test has good power even when the degree of causality c is low. Besides, the S_T has small size distortions when the sample size is small and the correct asymptotic size when

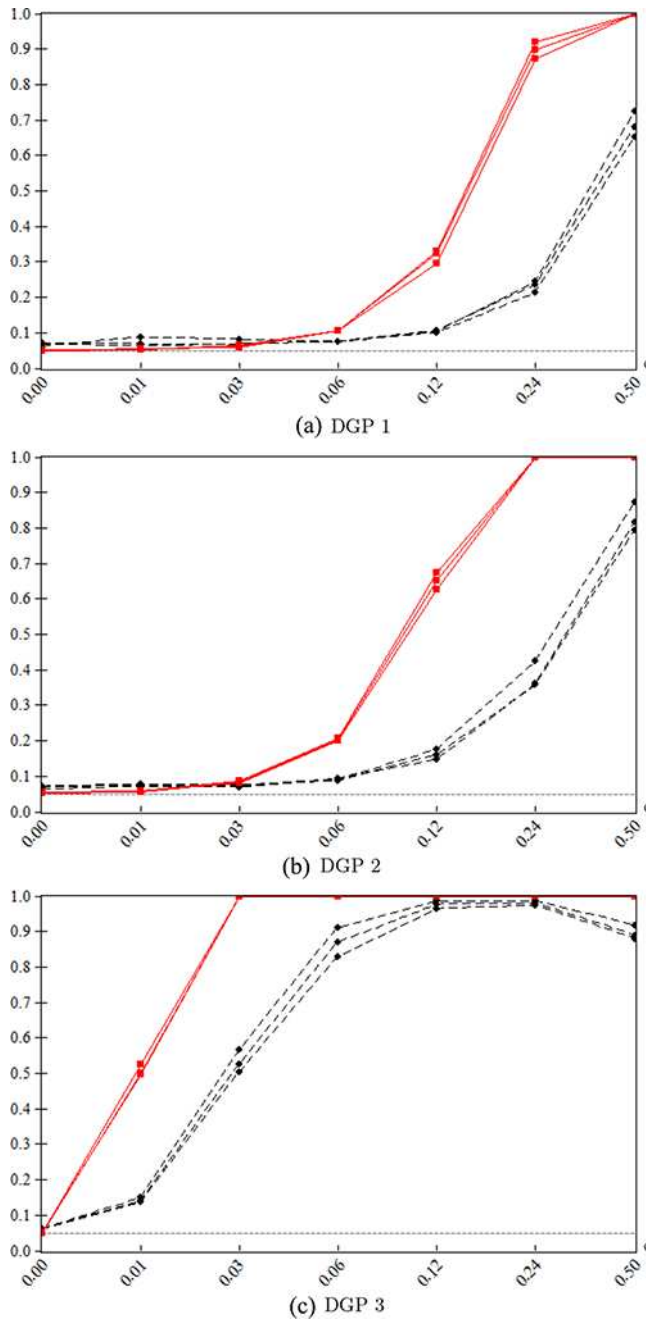


Figure 1. Empirical rejection frequencies for 5% subsampling S_T test—QAR(1) model of (17) with subsample sizes of $b_k = \lfloor kT^2/5 \rfloor$, $k = \{3, 4, 5\}$ - $T = 100$ (---), and $T = 500$ (—).

¹We do not include the results for $T = 250$ to save space.

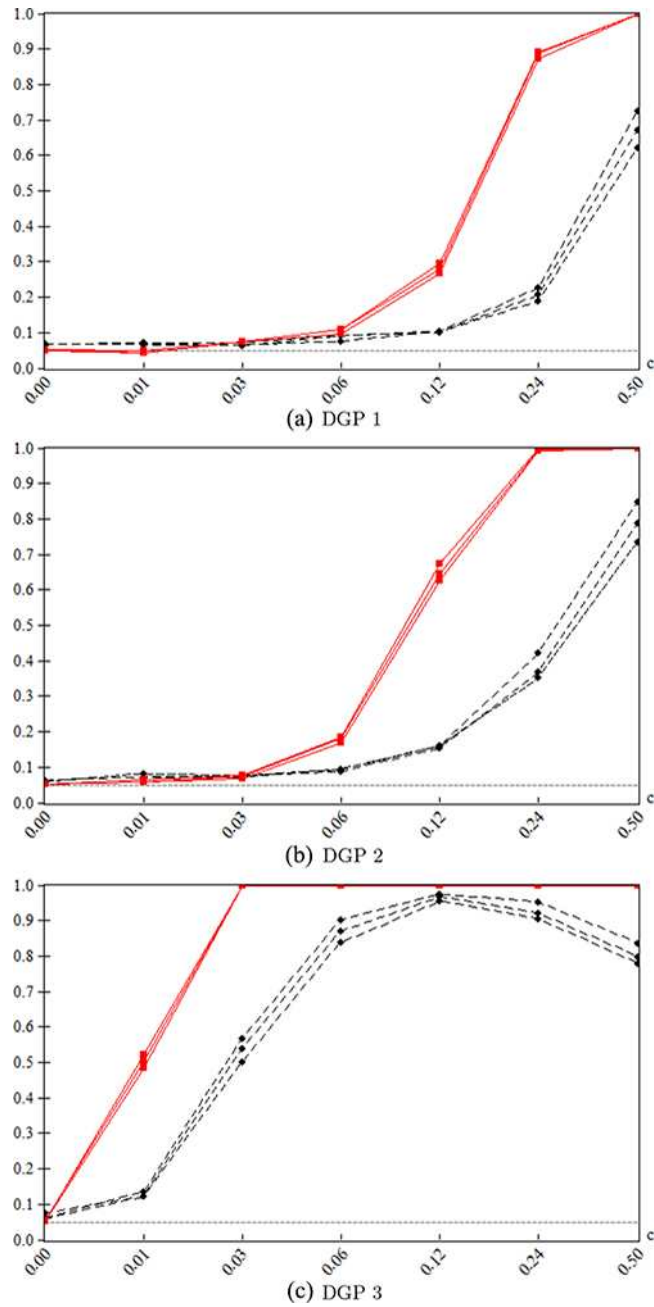


Figure 2. Empirical rejection frequencies for 5% subsampling S_T test—QAR(2) model of (17) with subsample sizes of $b_k = \lfloor kT^{2/5} \rfloor$, $k = \{3, 4, 5\}$ - $T = 100$ (---), and $T = 500$ (—).

$T = 500$. As DGP3 is presented in Jeong et al. (2012), Figs. 1 and 2 show that our test not only outperforms their test for $T = 500$, but also that it obtains reliable results for a smaller sample size of $T = 100$. Figure 3 displays the test results for increasing ratios of the coefficients of Y_{t-1} and Z_{t-1} to the variance of the error term. The test statistic has a reliable size and considerable power even for small ratios of the coefficients of Y_{t-1} and Z_{t-1} to the variance of the error term. Thus, for different DGPs and conditional quantile regression models, the S_T has correct asymptotic size and an increasing power with the sample size.

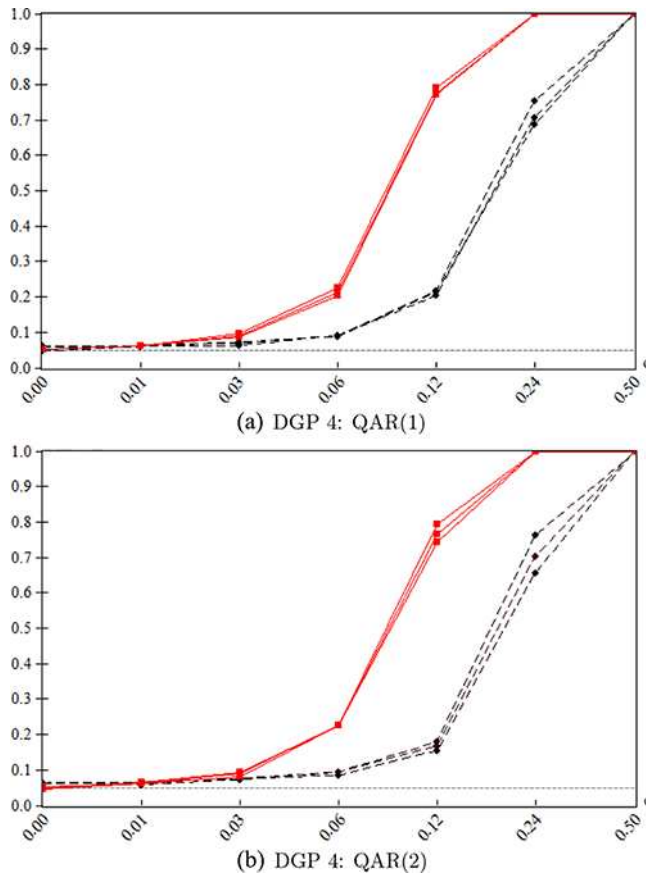


Figure 3. Empirical rejection frequencies for 5% subsampling S_T test—increasing ratios of Y_{t-1}, Z_{t-1} to the variance of the error term with subsample sizes of $b_k = \lfloor kT^{2/5} \rfloor, k = \{3, 4, 5\}$ - $T = 100$ (---), and $T = 500$ (—).

We also compare our results with the Sup-Wald test statistic proposed by Koenker and Machado (1999). To calculate the Sup-Wald test statistic, we include the variable Z_{t-1} on the linear conditional quantile regression models in (17). Without loss of generality, we assume that each quantile regression model is correctly specified if we include Z_{t-1} in the quantile regression model. Then we consider the following parametric specifications for the quantiles of Y_t :

$$\begin{aligned} \text{W1: } m_*^1(I_t^Y, I_t^Z, \theta_T(\tau)) &= \mu_0(\tau) + \mu_1(\tau)Y_{t-1} + \beta_1(\tau)Z_{t-1} + \sigma_t\Phi_\varepsilon^{-1}(\tau), \\ \text{W2: } m_*^2(I_t^Y, I_t^Z, \theta_T(\tau)) &= \mu_0(\tau) + \mu_1(\tau)Y_{t-1} + \beta_1(\tau)Z_{t-1} + \mu_2(\tau)Y_{t-2} + \sigma_t\Phi_\varepsilon^{-1}(\tau). \end{aligned} \quad (18)$$

Given a conditional linear quantile regression model in W1-W2 above, testing the null hypothesis of Granger noncausality by the Sup-Wald test consists in testing $H_0^{SW} : \beta_1(\tau) = 0$, for all $\tau \in \mathcal{T}$. Figure 4 gives the results for the Granger-causality tests based on the Sup-Wald test. A drawback of the Sup-Wald test is that the critical values do not have the correct nominal size in small samples, as the empirical sizes are always smaller than the 5% nominal level of the test. The results also suggest that the subsampling S_T test considerably outperforms the Sup-Wald procedure in terms of power. For the DGPs considered, even using a small subsample size, $b = \lfloor 3T^{2/5} \rfloor$, the S_T presents powerful and reliable inference. In addition, the subsampling S_T test is robust to the choice of the subsample size. In sum, our proposed test seems to perform quite well in finite samples.

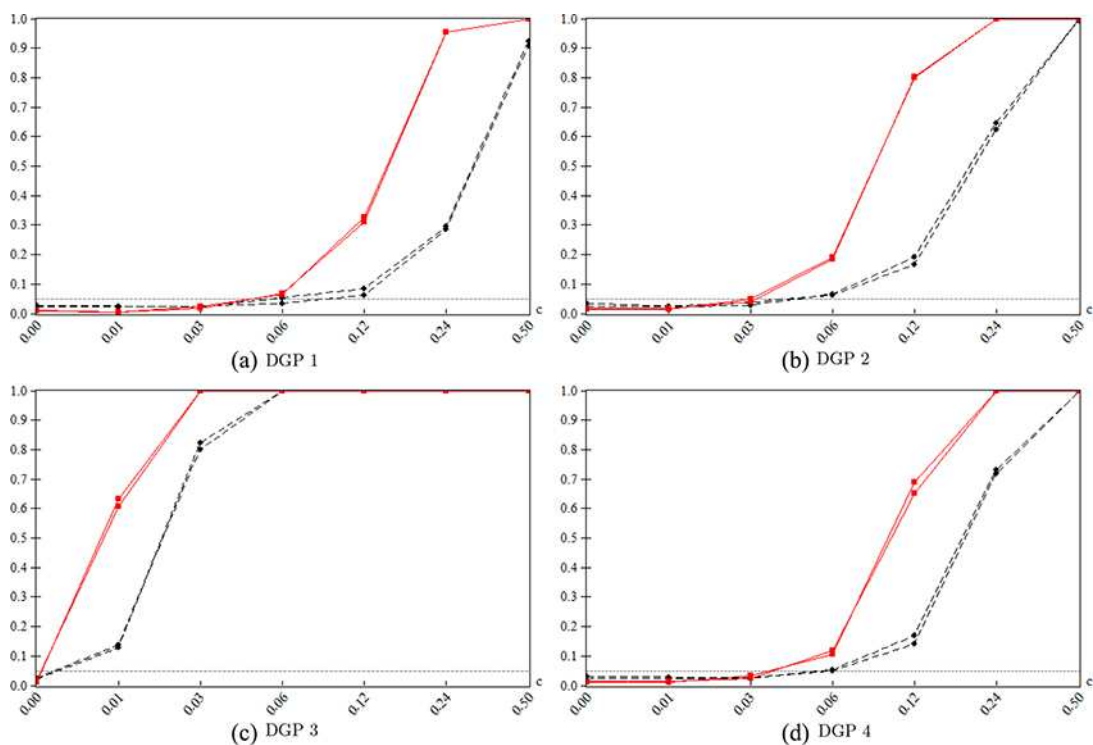


Figure 4. Empirical rejection frequencies for 5% Sup-Wald test—quantile models W1-W2 of (18) - $T = 100$ (--) and $T = 500$ (—).

5. Empirical application

To illustrate the applicability of our approach, we analyze the causality between gold prices, oil prices, and USD/GBP exchange rate. The gold and oil market are the main representatives of the large commodity markets. Gold is a valuable asset and can maintain its value in turbulent times. There is evidence that the gold and oil markets have a close interaction. For instance, both gold and crude oil prices entered into a boom time in 2002 due to US dollar depreciation, global inflation, and oil supply manipulation by the Organization of the Petroleum Exporting Countries (OPEC); and both commodity prices collapsed together in the financial crisis of 2008 (Zhang and Wei, 2010). Therefore, it is important to study how gold and oil prices vary, and their causal relationship. While standard tests evidence a positive mean causal relation between oil and gold prices, our main goal is to evaluate such a relation on each quantile of the distribution.

We apply our S_T test to check the relationship between the S&P gold prices (per ounce) and Brent crude oil prices (per barrel). Under our approach, we can discriminate between causality affecting the median and the tails of the conditional distribution. Then the empirical analysis should provide a more complete description of the causal relation between gold and oil prices. We also evaluate the effect of the USD/GBP exchange rate on gold prices to compare the performance of our parametric test with the nonparametric approach proposed by Jeong et al. (2012). The data consist of 3,440 daily observations—from July 2000 to September 2013—on the USD/GBP exchange rate, oil and gold prices, and all series were obtained from Datastream.

Figure 5 displays the daily log and log-difference series. The graphs of the log series indicate that the series are nonstationary and follow a common pattern. Table 1 presents the summary statistics of the series. The gold and oil prices are very volatile, and all series are positively skewed and leptokurtic. We apply the S_T test on the log-difference of the series, as Dickey–Fuller and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) unit root tests show that the three log series are nonstationary.

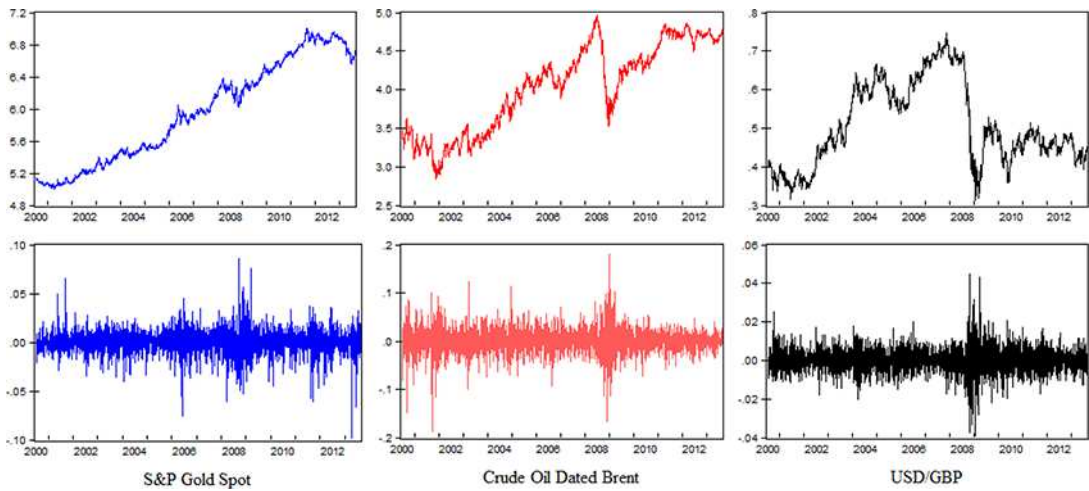


Figure 5. Log of the series (upper panel) and first-difference of the logs (lower panel).

Table 1. Summary statistics.

	Gold prices	Oil prices	USD/GBP
Mean	466.89	65.31	1.67
Std. Dev.	282.83	33.23	0.18
Median	382.74	62.06	1.61
Skewness	0.64	0.32	0.49
Kurtosis	2.04	1.81	2.05
Minimum	149.39	17.00	1.37
Maximum	1101.48	143.60	2.11

Gold is the S&P GSCI Gold Spot price index; Oil price is the price adjusted default Crude Oil Dated Brent in US dollars per barrel; USD/GBP is the exchange rate of US dollars to UK British pounds; The data covers the period that spans July 3, 2000 to September 6, 2013.

In order to compare our approach with a standard test of Granger-causality, we perform a Granger-Causality in mean between the differences of the log of the three series. We test the null hypothesis of Granger noncausality in mean of (2), for three different lag specifications. Table 2 presents the p -values of the Granger-Causality test between ΔGold , ΔOil , and $\Delta\text{USD/GBP}$. There is no Granger-causality in mean from $\Delta\text{USD/GBP}$ to ΔGold and to ΔOil , at the 5% significance level, for different lag specifications. Besides, we find no Granger-causality from variations in oil prices to variations in gold prices, at the 5% significance level. However, gold price variations Granger-cause in mean oil price variations at the 5% significance level. Table 2 displays no pattern of Granger-causality in mean at the 1% significance level. In sum, there is no evidence of a strong pattern of Granger-causality between the series using a conditional mean analysis.

To apply our S_T test of (6), we estimate three different quantile auto-regressive (QAR) models as in (17) for each dependent variable on the S_T test. Tables 3 and 4 report the subsampling p -values of our S_T test. If we take into account all quantiles, the results suggest that variations in the gold prices Granger-cause variations in the oil prices, and vice-versa, at the 1% significance level. However, if we perform a median-regression and consider only $\tau = 0.50$, there is no evidence that variations in the oil prices Granger-cause variations in the gold prices, and vice-versa, at the 5% significance level.

In addition, changes in the USD/GBP Granger-cause changes in the gold and oil prices at the 1% significance level, when we consider all quantiles of the conditional distribution. However, if we consider only the extreme tails of the conditional distribution for $\tau = 0.10$ and $\tau = 0.90$, we often cannot reject that variations of USD/GBP do not Granger-cause variations in gold prices and in oil prices, at the 1% significance level. These results are consistent with the ones obtained by Jeong et al. (2012), who showed that USD/GBP exchange rate changes do not cause the gold price change if $\tau < 0.22$ or $\tau > 0.80$.

Table 2. Granger-causality in mean: p -values.

T = 3,440	Number of Lags		
	1	2	3
<i>A. Granger-Causality in Mean to ΔGold</i>			
Δ Oil to Δ Gold	0.314	0.548	0.097
Δ USD/GBP to Δ Gold	0.965	0.749	0.821
<i>B. Granger-Causality in Mean to ΔOil</i>			
Δ Gold to Δ Oil	0.017*	0.014*	0.035*
Δ USD/GBP to Δ Oil	0.421	0.752	0.154

* denotes significance at the 5% level.

Table 3. Granger-causality to Δ Gold: subsampling p -values.

T = 3,440	τ	Number of Lags of I_t^Y		
		1	2	3
Δ Oil to Δ Gold	[0.10; 0.90]	0.000**	0.000**	0.000**
	0.10	0.000**	0.000**	0.000**
	0.50	0.294	0.265	0.323
	0.90	0.000**	0.000**	0.000**
Δ USD/GBP to Δ Gold	[0.10; 0.90]	0.000**	0.000**	0.000**
	0.10	0.011*	0.010*	0.006**
	0.50	0.004**	0.006**	0.007**
	0.90	0.010*	0.007**	0.013*

* and ** denote significance at the 5 and 1% levels, respectively; For $\tau \in [0.10, 0.90]$.

Table 4. Granger-causality to Δ Oil—subsampling p -values.

T = 3,440	τ	Number of Lags of I_t^Y		
		1	2	3
Δ Gold to Δ Oil	[0.10; 0.90]	0.000**	0.000**	0.000**
	0.10	0.000**	0.000**	0.000**
	0.50	0.461	0.345	0.384
	0.90	0.000**	0.000**	0.000**
Δ USD/GBP to Δ Oil	[0.10; 0.90]	0.000**	0.000**	0.000**
	0.10	0.011*	0.010*	0.006**
	0.50	0.005**	0.006**	0.006**
	0.90	0.010*	0.006**	0.012*

* and ** denote significance at the 5 and 1% levels, respectively; For $\tau \in [0.10, 0.90]$.

In contrast to the mean analysis of Table 2, our S_T test detects a pattern of Granger-causality in the tails of the distribution from Δ Oil to Δ Gold (Table 3) and of Granger-causality in quantiles from Δ USD/GBP to Δ Gold and to Δ Oil (Tables 3 and 4). Therefore, the empirical application shows the ability of our approach to detect tail and quantile Granger-causalities that could be possibly ignored under a conditional mean regression analysis.

6. Conclusions

Many important policy and financial analyses are investigated through testing for Granger-causality between economic time series. However, most of the results in the literature were obtained in the context of Granger-causality in mean. In this article, we present a consistent parametric test of Granger-causality in quantiles. Rather than focusing on a single part of the conditional distribution, we develop a test that evaluates possible causal relations in all conditional quantiles. The proposed test statistic has correct asymptotic size, is consistent against fixed alternatives, and has power against Pitman deviations from the null hypothesis. In addition, the proposed approach allows us to evaluate nonlinear causalities, causal relations in conditional quantiles, and provides a sufficient condition for Granger-causality when all quantiles are considered.

Finite sample experiments suggest that our proposed test has good size and power properties, and is more powerful than other comparable test in the literature against almost all alternatives. An empirical application highlights the practical importance of our setting considering the causal relation between the gold price, the USD/GBP exchange rate, and the oil price. We illustrate that oil price, USD/GBP, and gold price changes presented a different causal relationship in the tail and in the center of the distribution.

A possible direction for future work is to extend this method to analyse the effect of misspecifications in the quantile regression model to Granger-causality. A possibly misspecified quantile regression model may lead to over-rejections of the Granger noncausality null hypothesis.

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